IDEA SUMMARY - Persuasively describe your idea briefly and succinctly. (1300 characters or spaces-with spaces)

Mindbridge is a software-first assistive technology that maps subtle human micro-signals into physical actions, initiated by assistive devices. Millions of people who may have developed certain neurological conditions face a situation in which they know exactly what they want to do, but their bodies struggle to consistently execute those movements. Small pressure signals are captured from a wearable wristband, and MindBridge translates them into universal digital commands that can be sent to a variety of assistive technologies, such as robotic gloves. Designed to help individuals with Multiple Sclerosis, the system takes user-specific pressure patterns, runs them through an algorithm that maps them to a list of calibrated commands and blocking for noise, turning it into output that maps the user's intent to action. By creating a level of separation between the intelligence and hardware, MindBridge effectively serves as a universal compatibility layer that can be connected to a wide ecosystem of assistive devices. Universal compatibility reduces cost, allowing the system to scale across a litany of conditions and applications. The result is a scalable and affordable solution, empowering users to restore independence and interact with technology on their own terms.

Mindbridge is a software-first assistive technology that is able to look for the nuances in human action, especially the subtle micro-actions in individuals who experience a limited range of motion (ROM). This is of enormous importance currently with millions of individuals across American society alone who have developed various neurological conditions, causing their mobility to be reduced. Combating this very problem. Mindbridge can capture small variations in pressure across appendages on an attached wristband. Mindbridge’s prosperity technology translates these readings into digital commands that can interact with a multitude of assistive technologies, such as prosthetics, robotic appendages, and internal devices that may facilitate movement. Made specifically for individuals with Multiple Sclerosis (MS), our programs can take specific pressure signals that can parse through an expansive list of commands and create the warranted motion. Additionally, our program is able to recognize static inputs from stimuli that may not be from the user. Our technology effectively adjusts to these differences, capturing only the user's motions. In regard to cost efficiency, our universal compatibility allows for seamless integration and ensures a scalable expansion, allowing our technology to be implemented across the world, working against the increasing presence of neurological conditions . Moreover, it ensures the affordability of assistive technology for the individuals who need it, especially in a time where treatment and assistive technology are at an all-time high.

Mindbridge is a software-first assistive technology.

PROBLEM - Define the problem or opportunity you aim to address. (500 characters or spaces-with spaces)

Millions of Americans live with disabilities every day: specifically, 61 million of them. 5.5 million of those have effectively lost limb function. 1 million Americans are affected by conditions such as multiple sclerosis, which severs the link between human intention and the actuated movement. Typically, rehabilitation against these life degradations is infrequent, clinic-bound, and extremely expensive. As the number of patients increases, a scalable and universal solution is required.

According to JAMA Neurology, 180 million Americans are currently diagnosed with a neurological condition, ranging from headaches to strokes or even dementia-related conditions, resulting in the affliction of nearly 54% of the population. Each one of these individuals experiences some form of hindrance in their daily activities, with many losing some form of motion. Our technology will combat this issue head-on, rather than suggest expensive rehabilitation or therapy where they may or may not recover from their condition

SOLUTION - Describe specifically how your solution addresses the problem described above. (500 characters or spaces-with spaces)

Mindbridge is a translation platform linking small muscle pressure signals generated by the user to numerous assistive devices, converting human intent into deliberate movements. A wearable wristband detects pressure patterns, where it is sent to the proprietary MindBridge app. The app interprets the signals using adaptive algorithms, tailored and calibrated to the user and their known command patterns; this synthesis is sent to a connected output device, such as an actuated glove.

MindBridge is a software-first intent translation platform that converts deliberate muscle pressure signals into commands for assistive technologies. A wearable band detects intentional pressure patterns; the MindBridge app amplifies and interprets them using adaptive algorithms; the output activates devices like an actuated glove. Device-agnostic by design, MindBridge integrates across assistive hardware—enabling consistent, user-controlled rehabilitation outside clinical settings.

Mindbridge looks to challenge the problem head-on. Rather than falling into the norm of current technologies, which act passively, not confronting the reduced range of motion head-on. Mindbridge looks to work in real time, where it converts the user’s wishes into real-world consequences. Moreover, Mindbridge's ability to seamlessly integrate with various assistive

technologies allow it to be applied across different neurological conditions rather than be limited to a single niche.

UNIQUE VALUE PROPOSITION - Describe the benefit your solution offers that is unique and superior to any other alternatives. (1000 characters or spaces-with spaces)

MindBridge is a universal interface, designed to work with all different forms of assistive tech, not just a single device. The lack of universal solutions has proliferated costs for patients, as they require very specific technology to meet their specific needs. As MindBridge acts as a compatibility layer between the intent and action, it performs the interpretation of the signals, allowing patients to use more general/universal assistive tech that is more affordable. Since the MindBridge intelligence lives in the software, it continuously upgrades itself through algorithm updates, learning from user feedback/calibration, and growing support for different types of assistive technology. The initial focus is on multiple sclerosis, but MindBridge will expand to assist with additional situations, such as stroke recovery, cerebral palsy, and ALS. The scalable nature of MindBridge starts with initial pilot testing and then is pushed globally to assistive tech providers and medical centers.

MindBridge introduces a fundamentally different paradigm in assistive technology: a universal intent interface, not a single device. Where most tools target specific hardware for one condition, MindBridge acts as a compatibility layer between human intention and assistive devices—translating voluntary muscle signals into commands any connected device can execute. Two core processes power the platform: signal amplification and intent conversion, ensuring even small muscle activity produces reliable, meaningful output. Because intelligence lives in software, MindBridge improves continuously through algorithm updates, calibration learning, and expanding device compatibility. Initial focus is multiple sclerosis, but the architecture scales to stroke recovery, cerebral palsy, ALS, and adaptive prosthetics. By separating the intelligence layer from the hardware layer, MindBridge builds a scalable ecosystem where assistive technology becomes more accessible, adaptable, and user-driven.

Currently, there is no mediative technology that exists within rehabilitative medicine. There are only two ends of the spectrum. One side is the diagnostic side, which involves the application of artificial intelligence in screening methods, while the other side is the use of prosthetics that enable movement. However, neither side has a tangible bridge to guarantee the restoration of these individuals' quality of life. However, Mindbridge does just that. It helps connect prosthetic devices, especially with its compatibility to prediagnostic conditions, ensuring that the quality of

life is never lost. Moreover, the technology utilizes never-before-seen abstraction in its infrastructure. It makes the most of a multi-modal nature and can utilize a variety of modalities, especially the wide range of programming languages that may be found in various prosthetic technologies. In regard to a wide range of integration, Mindbridge is not limited to multiple sclerosis itself. Its hyperparameters can be adjusted to fit the needs of a variety of individuals suffering under a wide range of neurological conditions like ALS, cerebral palsy, and dementia-related diseases.

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TARGET CUSTOMER(S) - Who specifically will need your solution most? How do you know they have the problem above? (500 characters or spaces-with spaces)

MindBridge targets MS patients with unstable hand motor control. MS patients retain intent and small movement, but lose the precision needed for consistent, exact movements. Unlike stroke treatment, MS symptoms change often, making it incredibly difficult to have reliable control. Thus, MindBridge's real-time, dynamic intent framework is a perfect fit for their needs. This would help the nearly 1 million Americans that live with MS, giving them a tool for support outside of clinical settings.

MindBridge targets MS patients with inconsistent hand motor control. Most retain cognitive intent and partial muscle activation but lose the reliability needed for precise, consistent movement. Unlike stroke recovery, MS symptoms fluctuate daily, making sustained control exceptionally difficult; this unpredictability makes MS patients ideal for a real-time adaptive intent system. Nearly 1 million Americans live with MS, many depending on tools that preserve independence outside clinical settings

Mindbridge targets individuals who are currently afflicted by MS, especially those who have lost a significant range of motion. These individuals are those who will need their motor coordination for their job, daily activities, or even for their quality of life. Due to other conditions, MS is a degenerative disease, meaning that forms of paralysis or neuron degeneration is irreversible and cause long-lasting damage. This makes Mindbridge especially essential in their treatment process, as it acts as an active bridge to their past activities, restoring a level of order that would be impossible if it were not for Mindbridge.

COMPETITIVE ADVANTAGE - Who are your competitors? What is your solution's competitive advantage over the competitors? (1000 characters or spaces-with spaces)

Currently, patient recovery tools focus mostly on restrained hardware, not adaptability through software. Some notable examples include SaeboFlex, who uses spring mechanics to assist hand opening. This is effective in sessions, but not in daily use. MindBridge interprets user intent directly, providing actions by clear signals, not passive mechanics. MyoPro, on the other hand, uses EMG to power an arm orthosis, but is rooted in confusing and complex proprietary hardware. We separate interpretation from hardware, which allows multiple devices to be compatible seamlessly without excess specialization. Neofect focuses on guided therapy and rehab, but not actual device signaling. MindBridge goes beyond sessions, turning intention into device control and buttressing autonomy outside the clinic. The pattern is clear. Most tools are constrained by rigid parameters, but MindBridge approaches this fundamentally differently by focusing on the interpretation layer that connects human and hardware, allowing for multi-use, improvement of current technology, and a comfortable experience for all users.

Most rehabilitation tools target fixed hardware—not adaptable control. SaeboFlex uses spring mechanics to assist hand opening; effective in sessions, but doesn't translate to daily interactive use. MindBridge interprets user intention directly—assistive actions driven by deliberate signals, not passive mechanics. MyoPro uses EMG to power an arm orthosis, but is complex, costly, and locked to proprietary hardware. MindBridge decouples interpretation from hardware, spanning multiple devices without a specialized system. Neofect centers on guided therapy and recovery—not real-world device interaction. MindBridge extends beyond sessions, converting intention into live device control and sustaining autonomy outside clinical settings. Where most assistive tools are isolated ecosystems, MindBridge operates at the interpretation layer between human intent and hardware—enabling interoperability, amplifying existing technologies, and letting users engage more naturally and reliably.

KEY RESOURCES - What are the key resources you will need to make your idea a reality? (650 characters or spaces-with spaces)

To realize MindBridge, we need engineering talents in complex signal processing and assistive hardware compatibility, hardware materials and expertise for device integration, and clinical partnerships with recovery providers for initial pilot and user testing. We have already built a working prototype—a wearable band, assistive glove platform, and an intent-to-action interaction system, providing consistent, user-controlled activation. Further development will expand algorithm precision, ensure high performance across various patient populations, and integrate various assistive technologies for different rehabilitation cases.

MindBridge requires engineering talent in advanced signal processing and assistive hardware integration, hardware expertise for device compatibility, and clinical partnerships with rehabilitation providers for validation and user testing. Our team has already built a working proof-of-concept—wearable band, assistive glove interface, and signal-to-motion feedback system demonstrating repeatable, user-controlled activation. Future development expands algorithm reliability, validates performance across diverse patient populations, and integrates the platform with additional assistive technologies deployed in real rehabilitation settings.

WHAT'S YOUR WHY? - Why do you personally want to develop this business idea? (1000 characters or spaces-with spaces)

MindBridge was born from personal experience—several team members have families affected by debilitating neurological conditions. Watching motor control loss strip independence from daily life shaped a shared conviction: technology should restore autonomy, not create barriers. MindBridge's mission is simple—to provide care for those who cannot provide it for themselves. Shreyas Korlapati's focus on AI and human-centered systems drove exploration of how software bridges human intention and machine response. Arjun Sargur's passion for robotics and engineering shaped the design of devices that respond naturally to human signals. Ansh Pathak's expertise in system architecture built reliable signal interpretation logic. Varun Koduri's interest in biomedical innovation kept the concept grounded in real rehabilitation needs. Together, these strengths converged around one purpose: building technology that restores dignity, independence, and control for individuals facing motor challenges.

Mindbridge was the result of experiences that each team member had to face in their personal life. Seeing family members lose their happiness and independence from detrimental neurological conditions forced them to live a life of monotony and sadness. Driven to change this dark reality, the members of Mindbridge looked to synthesize their strengths to create a functional product that would provide a future for those who were unable to provide it themselves. Shreya’s Korlapati familial experiences with MS and his extended experience with artificial intelligence propelled him to explore a unique solution to a problem that was so close to him. He would pair his passion for AI-driven systems with Arjun Sagur and Ansh Pathak's affinity for robotics and functional mechanics. However, Mindbridge understands the biological need and concepts which need to be bridged, which is why his expertise in biomedical engineering was essentially in ensuring that this technology had a real-world application.